Introduction To Industry 4.0 And Industrial Internet Of Things

Assignment-1

Number of questions: **15** **Total marks: 10 X 1= 15**

1. What is a transducer?

a. A device that converts energy from one form to another
b. A type of sensor that detects only energy
c. A digital communication device
d. A part of a computer processor

Answer: (a)

Justification: Please refer to Slide 3 of Lecture 1 of Week 1

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2. What does sensor accuracy refer to?

a. The range of the sensor
b. The smallest detectable input change
c. The correctness of the output compared to a standard value
d. The sensor's response time

Answer: (c)

Justification: Please refer to Slide 8 of Lecture 1 of Week 1

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3. What is an active sensor?

a. A sensor that can only sense physical changes
b. A sensor that operates without an external power source
c. A sensor that independently senses the input
d. A sensor that operates without any energy consumption

Answer: (c)

Justification: Please refer to Slide 16 of Lecture 1 of Week 1

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4. What does 'Actuator' primarily refer to in IIoT systems?

a. A device that stores data
b. A device that processes information
c. A device that converts signals into physical action
d. A device that enhances signal strength

Answer: (c)

Justification: Please refer to Slide 21 of Lecture 1 of Week 1

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5. Zigbee technology is popular for _____.

a. Long-range communications
b. High power consumption
c. Low-cost, low-power wireless mesh networks
d. High data rate transfer

Answer: (c)

Justification: Please refer to Slide 13 of Lecture 2 of Week 1

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6. Which of the following is true about 6LoWPAN?

a. It is used for high-power industrial networks
b. It enables IPv6 packets over low-power wireless networks
c. It focuses on high-speed ethernet
d. It's a standard for satellite communications

Answer: (b)

Justification: Please refer to Slide 17 of Lecture 2 of Week 1

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7. Near Field Communication (NFC) is typically used for:

a. Long-range satellite communications
b. High-power industrial networks
c. Short-range wireless interaction between electronic devices
d. Storing large amounts of data

Answer: (c)

Justification: Please refer to Slides 20 and 21 of Lecture 3 of Week 1

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8. Which of the following actuators work on the principle of compression and decompression?

a. Hydraulic Actuators
b. Thermal Actuators
c. Electric Actuators
d. Mechanical Actuators

Answer: (a)

Justification: Please refer to Page 109 of the book Introduction to IoT by S Misra, A Mukherjee, A Roy

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9. ISA 100.11a is significant in IIoT because it _____.

a. Provides high-speed internet connectivity
b. Focuses on wireless network technology for industrial automation
c. Is used for long-range satellite communications
d. Enhances Bluetooth signal strength

Answer: (b)

Justification: Please refer to Slide 8 of Lecture 3 of Week 1

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10. What is the primary function of IEEE 802.15.4 in IoT?

a. High-speed data transmission
b. Providing security algorithms
c. Establishing low-rate wireless personal area networks
d. Data storage and retrieval

Answer: (c)

Justification: Please refer to Slide 4 of Lecture 2 of Week 1

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11. Z-Wave technology is best suited for _____.

a. Industrial communication
b. Home automation
c. Satellite communication
d. High-speed internet

Answer: (b)

Justification: Please refer to Slide 3 of Lecture 3 of Week 1

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12. What is the maximum data rate possible in Wireless HART?

a. 2500 b/s
b. 2.5 mb/s
c. 250 kb/s
d. None of the above

Answer: (c)

Justification: Please refer to Slide 23 of Lecture 2 of Week 1

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13. QoS 2 in MQTT is also known as _____.

a. At most once
b. At least once
c. Exactly once
d. None of the above

Answer: (c)

Justification: Please refer to Slide 8 of Lecture 5 of Week 1

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14. CoAP works on Request/Response framework based on the _____ architecture.

a. UDP
b. TCP
c. SCTP
d. DCCP

Answer: (a)

Justification: Please refer to Slide 10 of Lecture 5 of Week 1

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15. Infra-red sensor is an example of _____.

a. Acoustic Sensor
b. Mechanical Sensor
c. Electrical Sensor
d. Optical Sensor

Answer: (d)

Justification: Please refer to Page 190 of the book Introduction to Industrial Internet of Things and Industry 4.0 by S Misra, C Roy, A Mukherjee

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